

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

QUIZ 4 SET B – SOLUTION
DURATION: 25 MINUTES

SUMMER SEMESTER, 2024-2025
FULL MARKS: 20

CSE 4205: Digital Logic Design

Programmable calculators are not allowed.

Answer **all 3 (three)** questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

ID: _____

1. Draw a three-bit bidirectional shift register. Clearly show your choice of flip-flops and label the circuit properly. The register must shift **right** when $M = 1$. Trace the data path for this mode.

5
(CO3)
(PO2)

Solution:

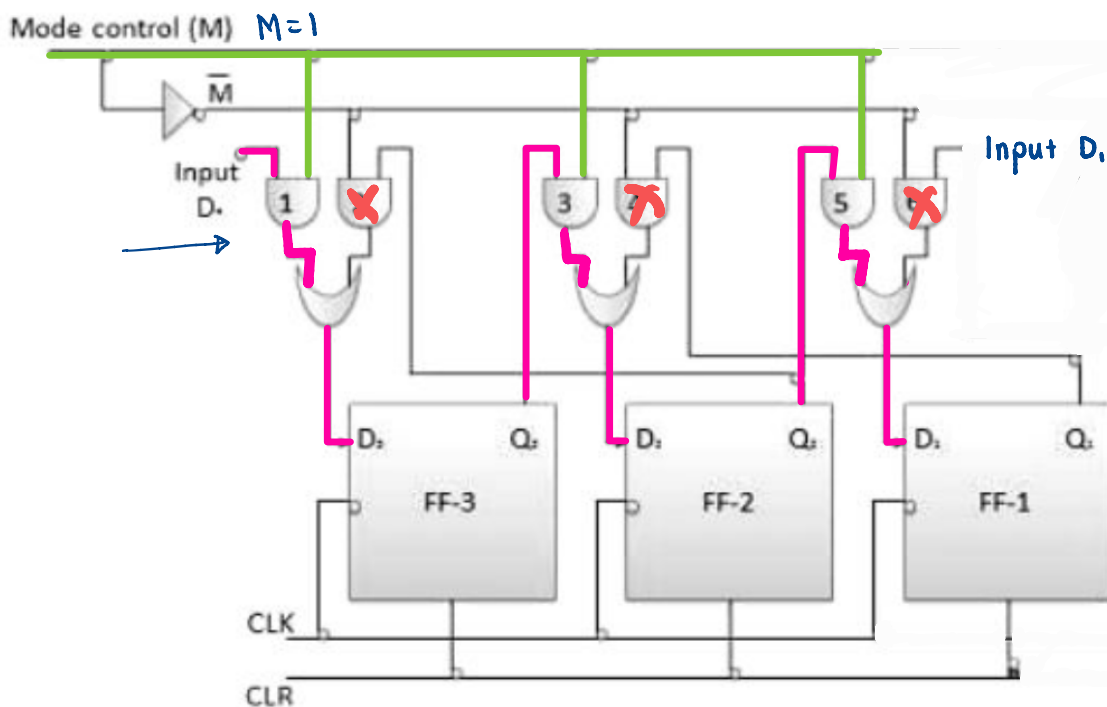
One valid design uses D flip-flops with input-selection logic. For $M = 1$ (right shift),

$$D_2 = MSR + \bar{M}Q_1, \quad D_1 = MQ_2 + \bar{M}Q_0, \quad D_0 = MQ_1 + \bar{M}SL.$$

Thus the active path is $SR \rightarrow Q_2 \rightarrow Q_1 \rightarrow Q_0$; the old Q_0 exits on the right.

Rubrics:

- 4 marks for a valid three-bit bidirectional design with clearly identified flip-flops, correct steering logic, labels, and mode selection.
- 1 mark for correctly tracing the $M = 1$ data path.



2. A Type-A flip-flop has two inputs and can exhibit the race-around condition for one input combination. A Type-B flip-flop has one input and is commonly used in counters and for controlled toggling. You must implement the behavior of Type B using Type A.

10
(CO3)
(PO2)

Complete the conversion using the following steps.

- | | | | |
|--|-----|--|-----|
| (a) Identify Types A and B. | [1] | (d) Construct the conversion table. | [2] |
| (b) Write the Type-B characteristic table. | [2] | (e) Derive the Type-A input equations. | [2] |
| (c) Write the Type-A excitation table. | [2] | (f) Draw the implementation. | [1] |

Solution:

Type A is a **JK flip-flop**; Type B is a **T flip-flop**.

Q	T	Q^+	Q	Q^+	J	K
0	0	0	0	0	0	X
0	1	1	0	1	1	X
1	0	1	1	0	X	1
1	1	0	1	1	X	0

Q	T	Q^+	J	K
0	0	0	0	X
0	1	1	1	X
1	0	1	X	0
1	1	0	X	1

Hence $J = T$ and $K = T$. Connect the T input to both J and K.

Rubrics:

- 1 mark for identifying both flip-flops.
- 2 marks for the correct Type-B characteristic table.
- 2 marks for the correct Type-A excitation table.
- 2 marks for the complete conversion table.
- 2 marks for the minimized Type-A input equations.
- 1 mark for the correct implementation diagram.

3. Design a three-bit synchronous sequential circuit using T flip-flops that cycles, in ascending order, through all three-bit values satisfying $x \bmod 2 = 1$, and then returns to the first value. Let the state be $Q_2Q_1Q_0$.

Complete the state/excitation table and derive minimized equations for T_2, T_1 , and T_0 . You do **not** need to draw the final logic circuit.

For reference, the T-flip-flop excitation table is

Q_n	Q_{n+1}	T
0	0	0
0	1	1
1	0	1
1	1	0

[Bonus: 2] After implementing your equations and connecting all flip-flops to the same clock, the circuit is powered up in state 000 and does not enter the required sequence. Explain why, and state one practical correction.

Solution:

The required sequence is

$$001 \rightarrow 011 \rightarrow 101 \rightarrow 111 \rightarrow 001.$$

Using $T_i = Q_i \oplus Q_i^+$ gives:

Present			Next			Excitation		
Q_2	Q_1	Q_0	Q_2^+	Q_1^+	Q_0^+	T_2	T_1	T_0
0	0	1	0	1	1	0	1	0
0	1	1	1	0	1	1	1	0
1	0	1	1	1	1	0	1	0
1	1	1	0	0	1	1	1	0

Treating unused states as don't-cares, the minimized equations are

$$T_2 = Q_1, \quad T_1 = 1, \quad T_0 = 0.$$

Bonus: The design equations specify only the required-state cycle. From 000, these equations produce the separate cycle

$$000 \rightarrow 010 \rightarrow 100 \rightarrow 110 \rightarrow 000,$$

so the circuit is not self-starting. Preset or parallel-load a valid state at reset, or explicitly assign unused-state transitions that enter the required cycle.

Rubrics:

- 3 marks for the correct required sequence and state/excitation table.
- 2 marks for the three minimized T-input equations.
- **Bonus 2:** 1 mark for identifying the invalid initialization/unused-state cycle and 1 mark for a valid correction.